SIEMENS

INSTALLATION INSTRUCTIONS
SIEMENS SL-SERIES 2-WIRE FIELD SELECTABLE HORN, STROBE, AND HORN/STROBE APPLIANCE (WALL MOUNT CLEAR AND AMBER LENS)

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PN P85580-001A

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Use this product according to this instruction manual. Please keep this instruction manual for future reference.

GENERAL

The Siemens LED based SL-Series SLHW horn, SLSW strobe, and SLHSW horn/strobe appliances are designed for easy installation. All models are for 24V operation. The SLHW horn is also for 12V operation. SLSW and SLHSW are for wall mount only. SLHW can be wall or ceiling mounted.

WARNING: Please read these instructions carefully before using this product. Failure to comply with any of the following instructions, cautions and warnings could result in improper application, candela setting, installation and/or operation of these products in an emergency situation, which could result in property damage and serious injury or death to you and/or others.

The Siemens LED based SL-Series meets NFPA 2016 20 millisecond light pulse duration code requirements. In addition, the Siemens LED based SL-Series product line has been UL/ULC listed as compatible with all Fire Alarm Control Panels (FACP) and accessories that have been determined to be compatible with Siemens model ST-Strobe based products including the ST, CH, SE, SEF, SET,ST,HS,MTH,S-HQ, SA, STH, ZR, and ZH Series. The maximum number of SL-Series devices per NAC is determined by divid¬ing the maximum current rating of the FACP NAC divided by the total appropriate current rating for the selected candela output of the SL-Series devices, with a maximum of 105 SL-Series devices per NAC. Refer to FACP installation instructions for more detail. The Siemens LED based SL-Series strobes may be installed in the same notification zone and field of view with any ST Strobe based product

Table 1: Specifications

Agency	SLSW, SLHSW: UL 1971, ULC-S526-16; SLSW-A: UL1638*, ULC-S526-16
	SLHSW, SLHW: UL 464: ULC-S525-16
Environmental	Indoor Use Only. 0° C -49°, C (32° F - 120° F) 93% R.H.
NAC Characteristics	Max. line resistance: 35Ω
Horn Patterns	Continuous, Code 3 (field selectable)
	Code 3 synchronized when using Siemens sync protocol
Horn Sound Output	High (HI), Low (LO) (field selectable)
Input Power	DC or FWR, 24V Regulated, 16 to 33V (All models)
	DC or FWR, 12V Regulated, 8 to 17.5V (SLHW only)
Strobe Candela	15, 30, 75, 110cd (field selectable)

*UL1638 is an on axis rating where the following applies:effective candela rating per UL1971;
Warning: Amber strobes are not to be used as a visual public mode alarm notification appliance.

STROBE AND HORN STROBE APPLIANCES

Siemens LED based SL-Series Multi-Candela Strobes can provide a non-synchronized strobe appliance when connected directly to a Fire Alarm Control Panel (FACP), or provide a synchronized strobe appliance when used in conjunction with an FACP that incorporates the Siemens sync protocol, a Dual Sync Module (DSC) , or the Siemens Power Supply.

NOTE: The Code 3 temporal pattern (1/2 second on, 1/2 second off, 1/2 second on, 1/2 second off, 1/2 second on, 1-1/2 off and repeat) is specified by ANSI and NFPA 72 for standard emergency evacuation signaling.

Table 2A: SLHW and SLHSW dBA Sound Output

Description		Reverberant dBA Per UL 464					
	Volume	SLHW at 12V			SLHW and SLHSW at 24V		
		8.0V	12.0V	17.5V	16.0V	24.0V	33.0V
Continuous Horn	High	80	84	87	80	83	86
	Low	78	79	84	78	81	81
Code 3 Horn	High	75	80	83	76	79	81
	Low	73	77	80	75	77	77

Table 2B: SLHW and SLHSW dBA Sound Output

Description		Anechoic Per CAN/ULC-S525-16					
	Volume	SLHW at 12V			SLHW and SLHSW at 24V		
		8.0V	12.0V	17.5V	16.0V	24.0V	33.0V
Continuous Horn	High	85	89	92	91	94	97
	Low	79	84	87	86	90	92
Code 3 Horn	High	85	89	92	91	95	96
	Low	79	84	87	86	90	91

Table 2C: ULC Directional Characteristics

-3dB	+ / -35 Degrees horizontal, +45 / -30 vertical
-6dB	+ / -90 Degrees horizontal, + / - 90 vertical

CURRENT DRAW:

Table 3: SLSW and SLSW-A Strobe Current Draw (Amps)

Strobe Candela Settings (cd)				
16.0-33.0 Volts				
Current	15	30	75	110
DC	0.030	0.040	0.102	0.201
FWR	0.040	0.059	0.149	0.267

Table 4: SLHSW Horn/Strobe Current Draw (Amps)

Strobe Candela Settings (cd)					
Current	Horn Settings	16.0-33.0 Volts			
		15	30	75	110
DC	High Continuous	0.042	0.052	0.110	0.217
	High Code 3	0.034	0.046	0.105	0.214
	Low Continuous	0.036	0.044	0.104	0.216
	Low Code 3	0.032	0.042	0.102	0.213
FWR	High Continuous	0.064	0.078	0.168	0.276
	High Code 3	0.052	0.069	0.159	0.269
	Low Continuous	0.059	0.077	0.163	0.273
	Low Code 3	0.047	0.064	0.158	0.267

NOTE: Candela and Horn Setting will determine the current draw of the product.

When calculating the total currents use Tables 3-5 to determine the highest value of RMS current for an individual appliance, then multiply these values by the total number of appliances. Be sure to add the currents for any other appliances, including audible signaling appliances powered by the same source, and to include any required safety factors.

Table 5: SLHW Horn Current Draw (Amps)			
Current	Horn Settings	8.0-17.5 Volts	16.0-33.0 Volts
DC	High Continuous	0.025	0.037
	High Code 3	0.024	0.030
	Low Continuous	0.020	0.026
	Low Code 3	0.018	0.022
FWR	High Continuous	0.051	0.059
	High Code 3	0.043	0.049
	Low Continuous	0.039	0.050
	Low Code 3	0.037	0.044

NOTE: These notification appliances are UL Listed as "Regulated". They are intended to be used with Fire Alarm Control Panels (FACPs) whose notification circuits are UL Listed as "Regulated." Refer to the FACP instructions or the Siemens Strobe Compatibility Data Sheet (PN 315-096363) for special application and strobe synchronization compatibility.

NOTE: These appliances were tested to the regulated voltage limits of 16.0-33.0 Volts for 24 volt models and 8.0-17.5 Volts for 12 volt models using filtered DC for the 12 volt range and either filtered DC or unfiltered DC for the 24 volt range voltage. Do not apply voltage outside of this range.

NOTE: Check the minimum and maximum output of the power supply and standby battery and subtract the voltage drop from the circuit wiring resistance to determine the applied voltage to the strobes. The maximum wire impedance between strobes shall not exceed 35 ohms.

NOTE: Strobes are not designed to be used on coded systems in which the applied voltage is cycled on and off.

NOTE: Make sure that the total rms current required by all appliances that are connected to the system's primary and secondary power sources, Notification Appliance Circuit (NAC), sync module, DSC sync modules, or Siemens power supplies does not exceed the power sources' rated capacity or the current ratings of any fuses on the circuits to which these appliances are wired.

WARNING:Overloading power sources or exceeding fuse ratings could result in loss of power and failure to alert occupants during an emergency, which could result in property damage and serious injury or death to you and/or others.

LIGHT OUTPUT:

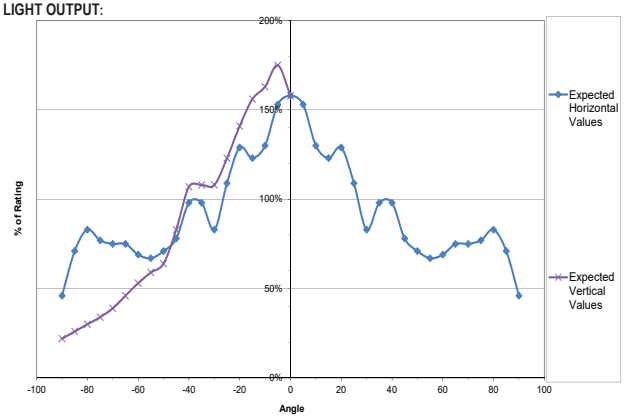


Figure 1: **SLHSW,SLSW and SLSW-A Expected Light Output**

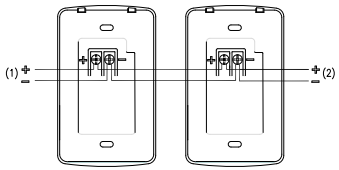


Figure 2 LED Strobe Wiring



Figure 3 **Wire Connection**

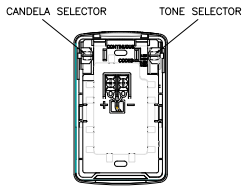


Figure 4: **Candela Selector**

WIRING, SETTINGS AND MOUNTING

• All strobe appliances have in-out wiring terminals that accept two #12 to #18 American Wire Gauge (AWG) wires at each screw terminal. Strip leads 3/8 inches and connect to screw terminals.

• Break all in-out wire runs on supervised circuits to ensure integrity of circuit supervision as shown in Figure 3. The polarity shown in Figure 2, the wiring diagram, is for the operation of the appliances. The polarity is reversed by the FACP during supervision.

NOTE: Wiring method shall be in accordance with CSA C22.1, Canadian Electrical Code, Part 1, Safety Standard for Electrical Installations,Section32.

NOTE: Do not fully back out terminal screws.

NOTE: The SLHSW and SLHW are factory set for the most common application of High dB and Code 3. The SLHSW and SLSW are set to 15 candela.

NOTE: Candela settings are shown in Figure 4.

MOUNTING OPTIONS

NOTE: Remove the Lens Protector Tape Before Replacing The Appliance Cover Grille.

1. Connect field wiring to contacts on back of device.
2. Dress wires back into backbox.
3. Install device as shown in Figure 5 to a single-gang backbox with the provided pan head screws.
4. Snap beauty cover over device.
5. To remove the appliance, insert a small flat-bladed screwdriver into the bottom opening ½" as shown in Figure 6. Then pry off beauty cover with the screw driver and then unscrew device.

NOTE: backbox must be recessed flush with the wall surface.

IMPORTANT: Device only has one mounting orientation. LED light element should be pointed towards ground.

NOTE: For other mounting options please use beauty plate (Red: SLSGP-R, White: SLSGP-W) or surface box (Red: SLSBBR, White: SLSBBW)

WARNING: DO NOT PAINT THIS DEVICE.

WARNING: When installing strobes in an open office or other areas containing partitions or other viewing obstructions, special attention should be given to the location of the strobes so that their operating effect can be seen by all intended viewers, with the intensity, number, and type of strobes being sufficient to make sure that the intended viewer is alerted by proper illumination, regardless of the viewer's orientation.

WARNING: A small possibility exists that the use of multiple strobes within a person's field of view, under certain circumstances, might induce a photo-sensitive response in persons with epilepsy. Strobe reflections in a glass or mirrored surface might also induce such a response. To minimize this possible hazard, Siemens strongly recommends that the strobes installed should not present a composite flash rate in the field of view which exceeds five (5) hz at the operating voltage of the strobes. Siemens also strongly recommends that the intensity and composite flash rate of installed strobes comply with levels established by applicable laws, standards, regulations, codes and guidelines.

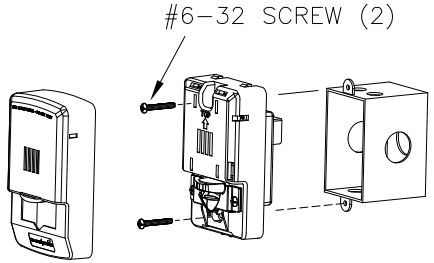


Figure 5: Installation

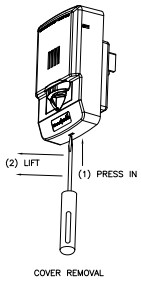


Figure 6: Removal (See step 5)

CAUTION: Check that the installed product will have sufficient clearance and wiring room prior to installing backboxes and conduit, especially if sheathed multiconductor cable or 3/4-inch conduit fittings are used.

Although the limits shown for the mounting option comply with the National Electrical Code (NEC) Siemens recommends use of the largest single gang backbox option available and the use of approved stranded field wires, whenever possible, to provide additional wiring room for easy installation and minimum stress on the product from wiring.

CAUTION: Do not over tighten mounting screws. Excessive torque can distort the base and may affect operation.

CAUTION: When using power tools to screw down the mounting plate to the electrical backbox, ensure the torque is set to the lowest setting available.

NOTE: NFPA 72/ANSI 117.1 conform to ADAAG Equivalent Facilitation Guidelines in using fewer, higher intensity strobes within the same protected area.

NOTE: Final acceptance is subject to Authorities Having Jurisdiction.

CAUTION: Check the installation instructions of the manufacturers of other equipment used in the system for any guidelines or restrictions on wiring and/or locating Notification Appliance Circuits (NAC) and notification appliances. Some system communication circuits and/or audio circuits, for example, may require special precautions to assure immunity from electrical noise (e.g., audio crosstalk).

NOTE: This equipment has been tested and found to comply with the limits for a Class A digital device, pursuant to part 15 of the FCC Rules. These limits are designed to provide reasonable protection against harmful interference when the equipment is operated in a commercial environment. This equipment generates, uses, and can radiate radio frequency energy and, if not installed and used in accordance with the instruction manual, may cause harmful interference to radio communications. Operation of this equipment in a residential area is likely to cause harmful interference in which case the user will be required to correct the interference at his own expense.

This Class A digital apparatus meets all requirements of the Canadian Interference-Causing Equipment Regulations.

Cet appareil numérique de la classe A respecte toutes les exigences du Réglement sur le matériel brouilleur du Canada.

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